PREDICTIVE ANALYTICS WITH ATHLETICS DATA

Week 12 · Introduction to Machine Learning Models

TUESDAY NOV 10 AND THURSDAY NOV 12, 2026

01

MACHINE LEARNING MODELS

MACHINE LEARNING MODELS

1 **Linear regression.**

Predicts a number from a weighted sum of the inputs. Weeks 5 and 6.

2 **Decision tree.**

Asks yes or no questions about the inputs to categorize the results

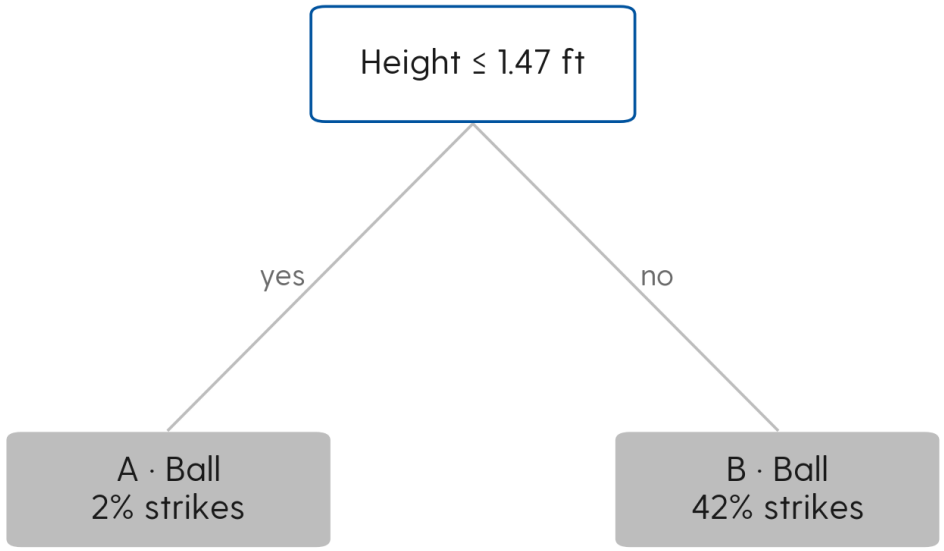
3 **Random forest.**

Mulitple decision trees, each with one vote



A BATTER TAKES A PITCH. WHAT WOULD YOU WANT TO KNOW ABOUT THE PITCH TO PREDICT THE CALL?

ONE LEVEL TREE STRIKE ZONE

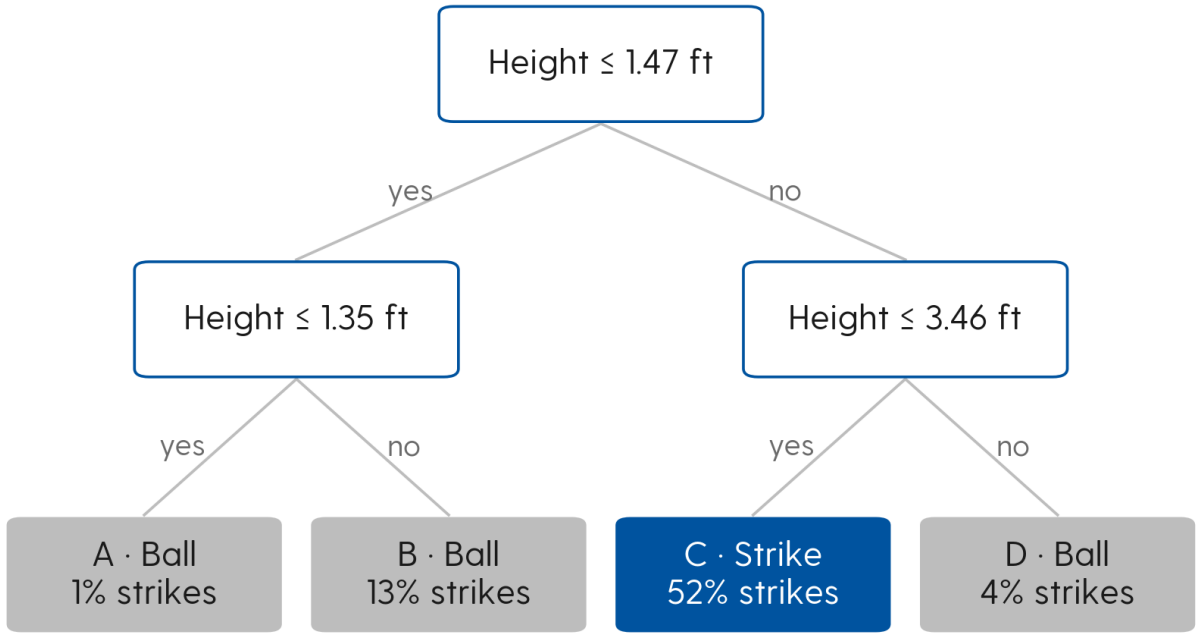


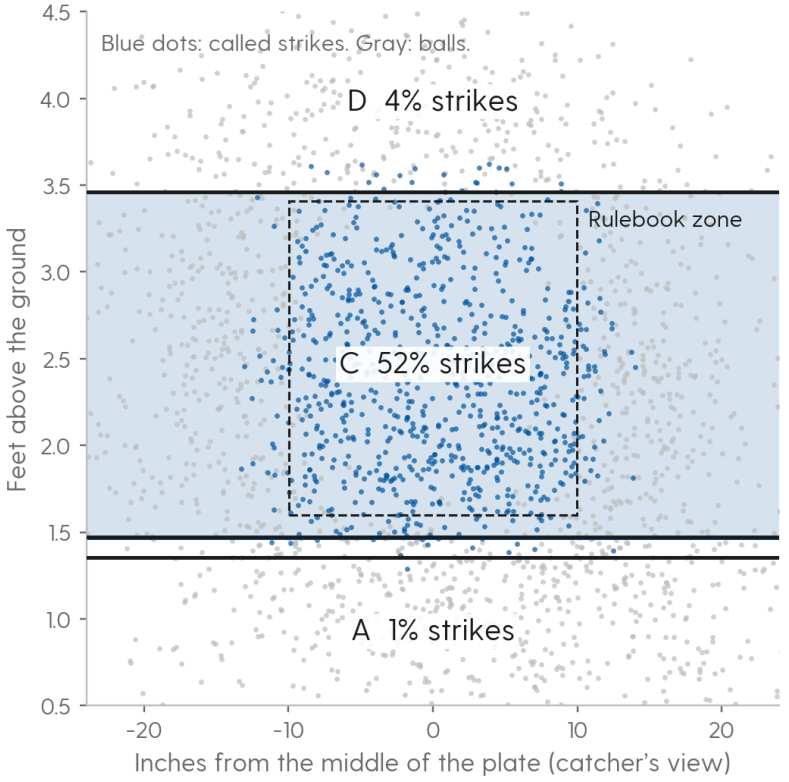


Every pitched is categorized as a ball. Accurate 68.3% of the time.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

TWO LEVEL TREE STRIKE ZONE

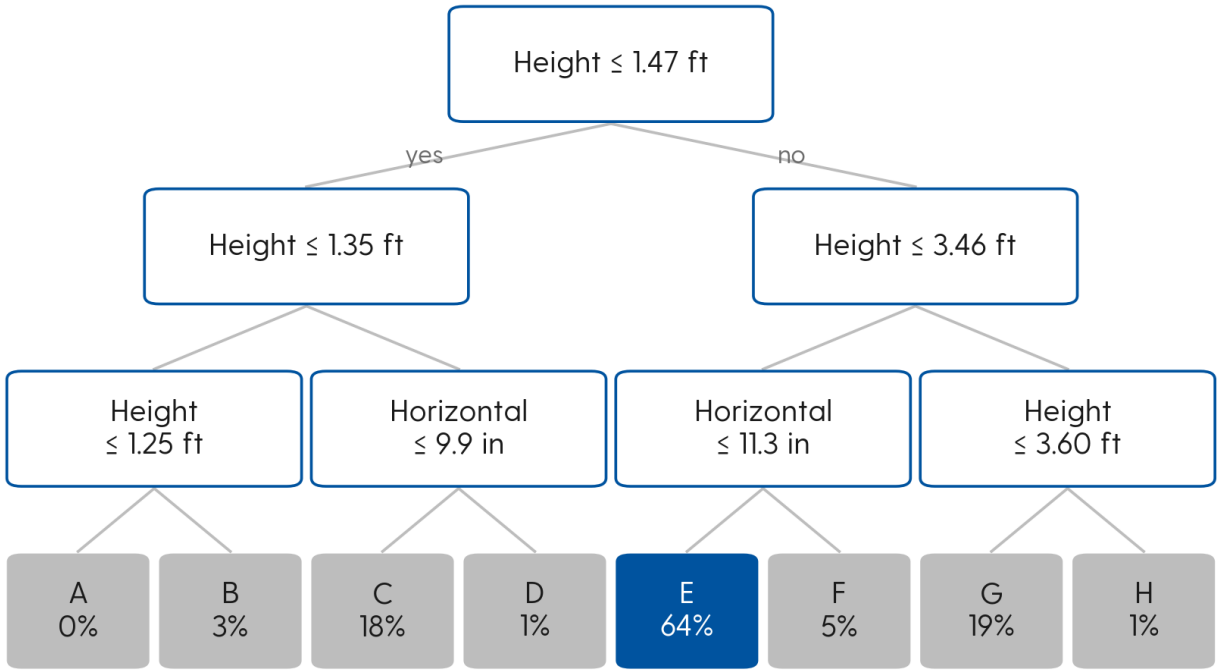


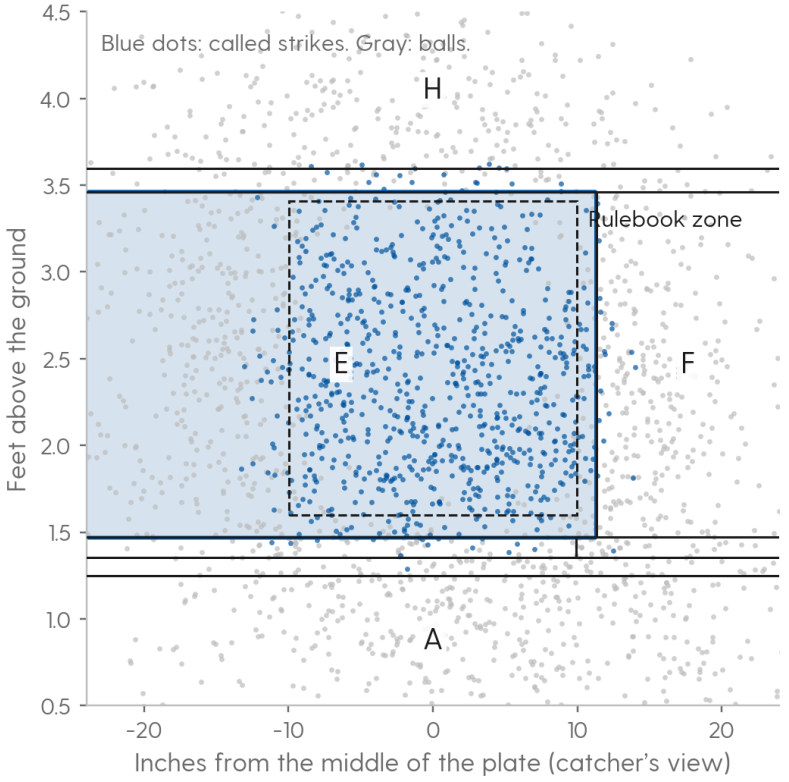


A band from the knees to the letters, with no sides: 70.8%.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

THREE LEVEL TREE STRIKE ZONE

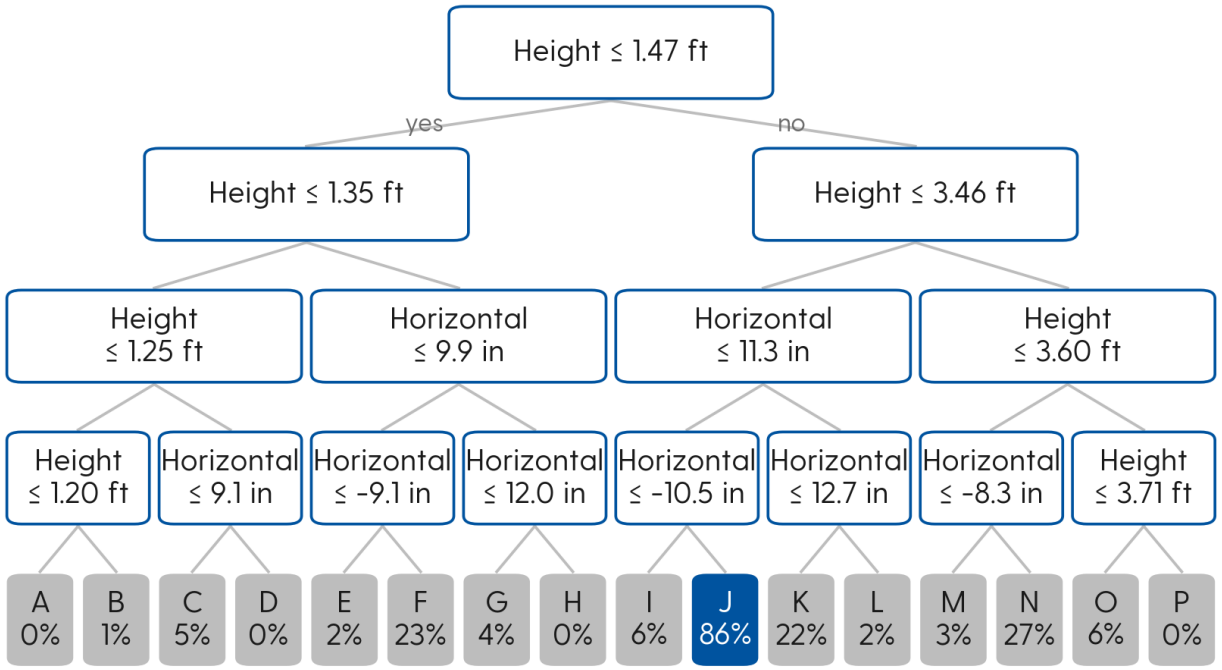


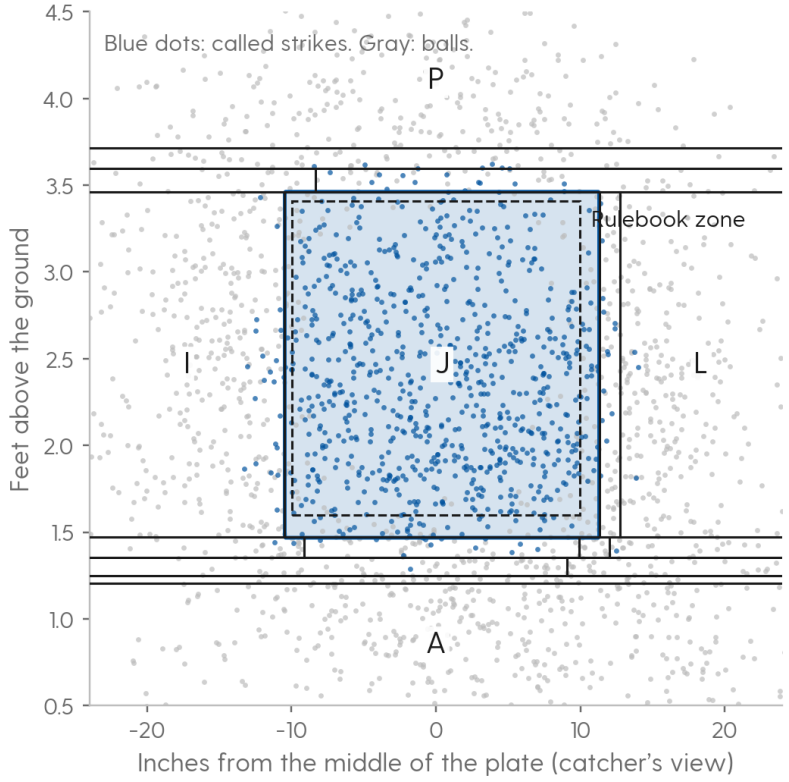


The third level finds the right edge of the plate, but not the left: 81.6%.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

FOUR LEVEL TREE STRIKE ZONE





Sixteen groups draw the zone: 92.3% of new pitches called right.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

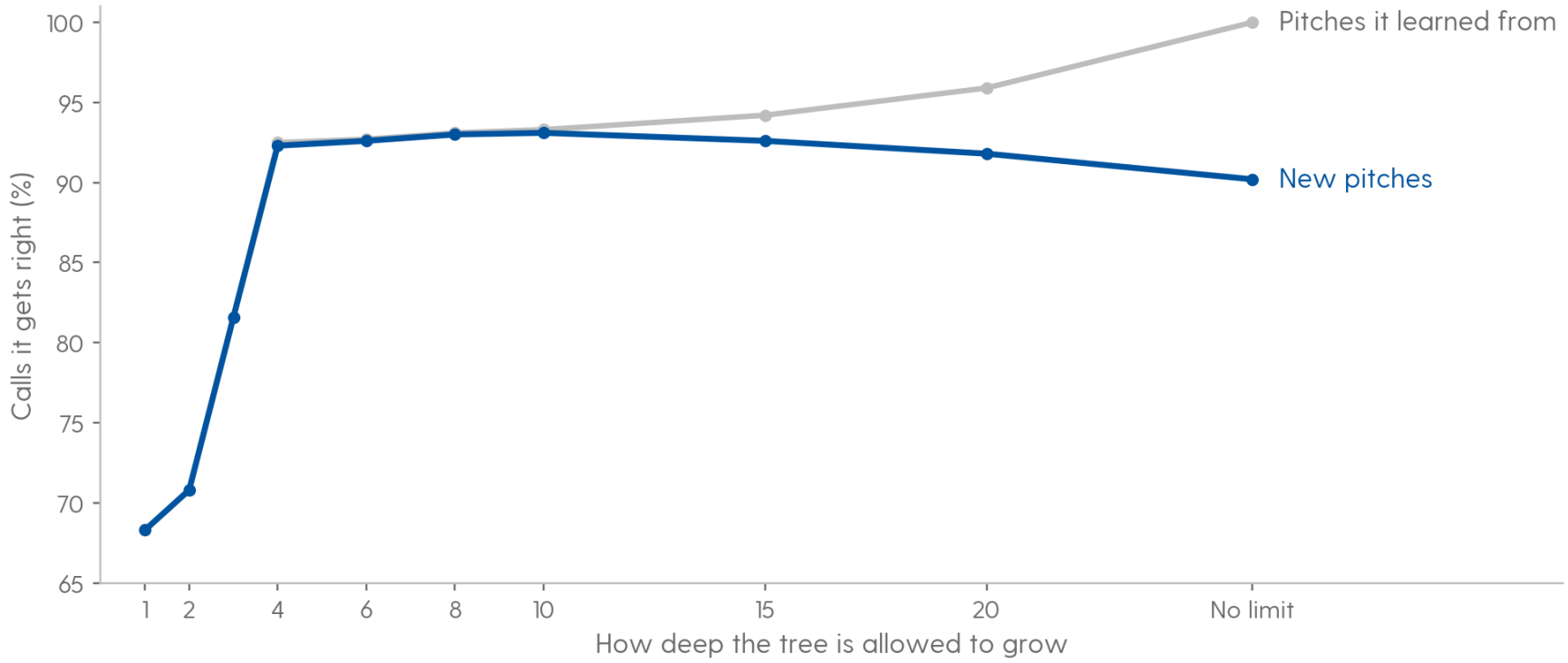
02

OVERFITTING



WHAT WOULD OUR ACCURACY LOOK LIKE IF THE TREE KEPT SPLITTING UNTIL EVERY GROUP WAS ONE ANSWER?

ACCURACY BY TREE DEPTH



The two lines match until about ten levels, then split. New pitches peak at 93.1% and fall to 90.2%.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

TRAIN/TEST SPLIT

Conducting a train/test split allows us to project how the model would perform on unseen data.

Typically, 80% of data is used for training, 20% for testing.

OVERFITTING MITIGATION

1. Increase dataset size.

2. Cross validation.

   Validate the model's accuracy by holding back portions of the training set. Use this process to determine the ideal hyperparameters, then test on unseen data.

3. Simplify the model.

   Manually tune hyperparameters (e.g.: limit the depth of a tree) or use a simpler model (e.g.: a ridge regression instead of a neural network.)

03

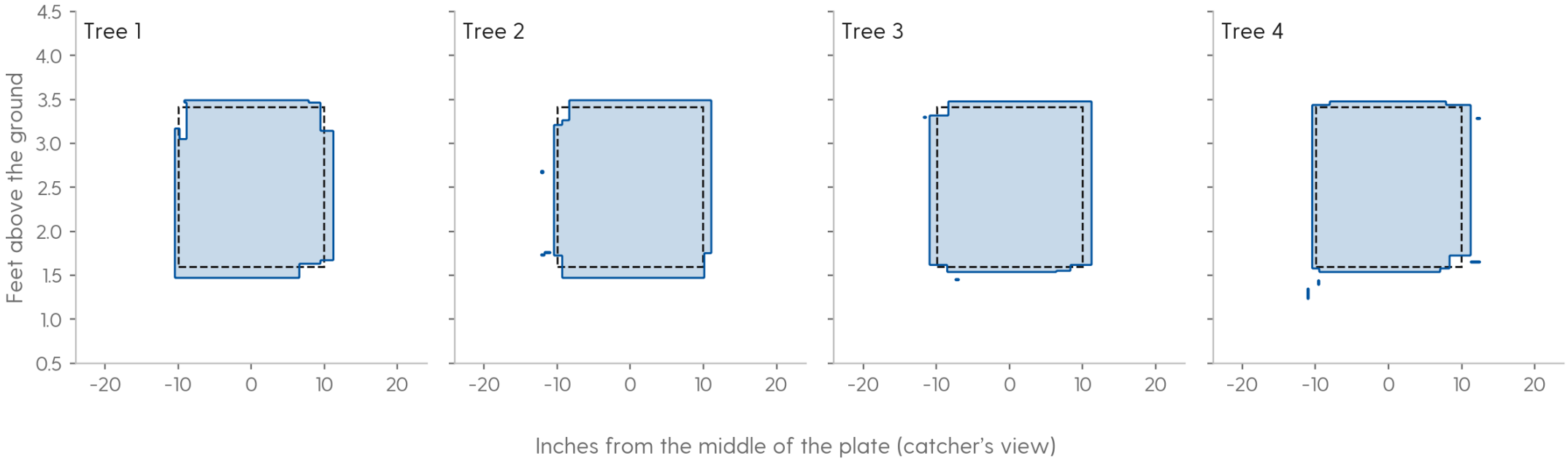
RANDOM FORESTS

RANDOM FOREST

1. Grow many trees, each on a random sample of the rows.
2. At each split, each tree considers a random subset of the columns.
3. The trees vote.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

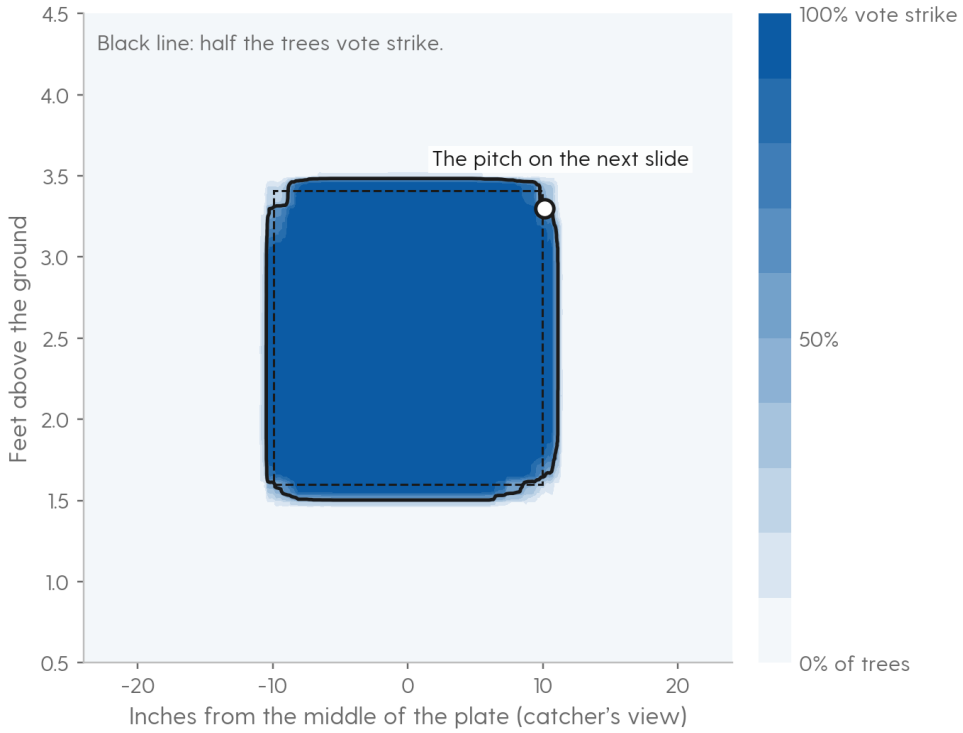
FOUR TREES FROM ONE FOREST



They agree in the middle of the zone and disagree at the edges.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

THE FOREST'S VOTE



Almost every tree agrees inside and outside the zone. Only a thin band at the edges splits the vote.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

A CLOSE CALL

	THIS PITCH
Where it crossed	10.1 in right of the middle, 3.30 ft high
Batter's zone	1.49 to 3.05 ft
Trees voting strike	60 of 100
Forest's call	Strike
One eight level tree's call	Ball
Umpire's call	Ball

The forest says how sure it is: 60 of 100 means a close call. A single tree only says yes or no.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

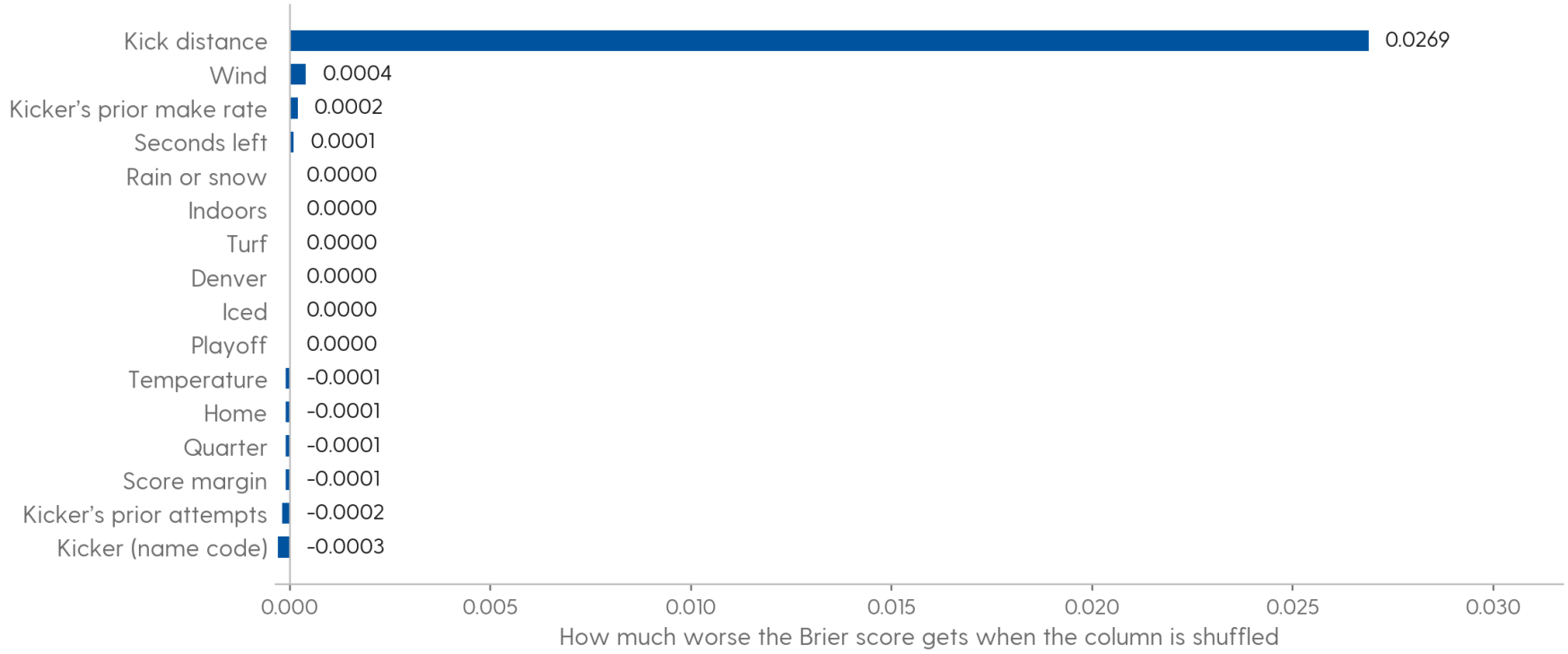
TREE VS FOREST

	SINGLE TREE	RANDOM FOREST
Can a coach read it?	Yes, a few yes or no questions	No, 100 trees
Accuracy on new pitches when deep	Drops when deep (90.2%)	Holds up better (92.2%)
Probabilities	A deep tree only says 0% or 100%	Any share from 0% to 100%
How you look inside	Read the questions	Permutation importance
Cost to train and run	One tree	100 trees

The forest predicts better. The tree is the one you can hand a coach.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

LOOKING INSIDE A FOREST



Shuffle kick distance and the forest's predictions get far worse. Shuffle any other column and they barely move.

Source: nflverse play-by-play, field goal attempts 2016 to 2025 (10,015 kicks).

04

CONFUSION MATRIX

? IS A PITCH A CALLED STRIKE? WHAT ARE THE TWO WAYS OUR MODEL CAN BE WRONG?

OPTIMIZATION GOALS

Accuracy

The model is right as often as possible. Typically the default optimization goal.

Precision

When the model predicts “positive” it’s usually correct. Minimizes false positives.

Recall

When something is actually “positive,” the model usually catches it. Minimizes false negatives.

WHAT SHOULD CANCER SCREENING BE OPTIMIZED FOR? SPAM EMAIL DETECTION? A FUMBLE VS. INCOMPLETE CALL?

Accuracy: The model is right as often as possible. Typically the default optimization goal.

Precision: When the model predicts “positive” it’s usually correct. Minimizes false positives.

Recall: When something is actually “positive,” the model usually catches it. Minimizes false negatives.

PENALIZING MISTAKES

1. A model learns by avoiding mistakes.

2. By default every mistake costs the same.

   When misses are rare, the cheapest answer is to say “good” every time.

3. You can make the mistake you care about cost more.

   The model trades some of the cheaper mistakes to avoid it.

05

CALIBRATION

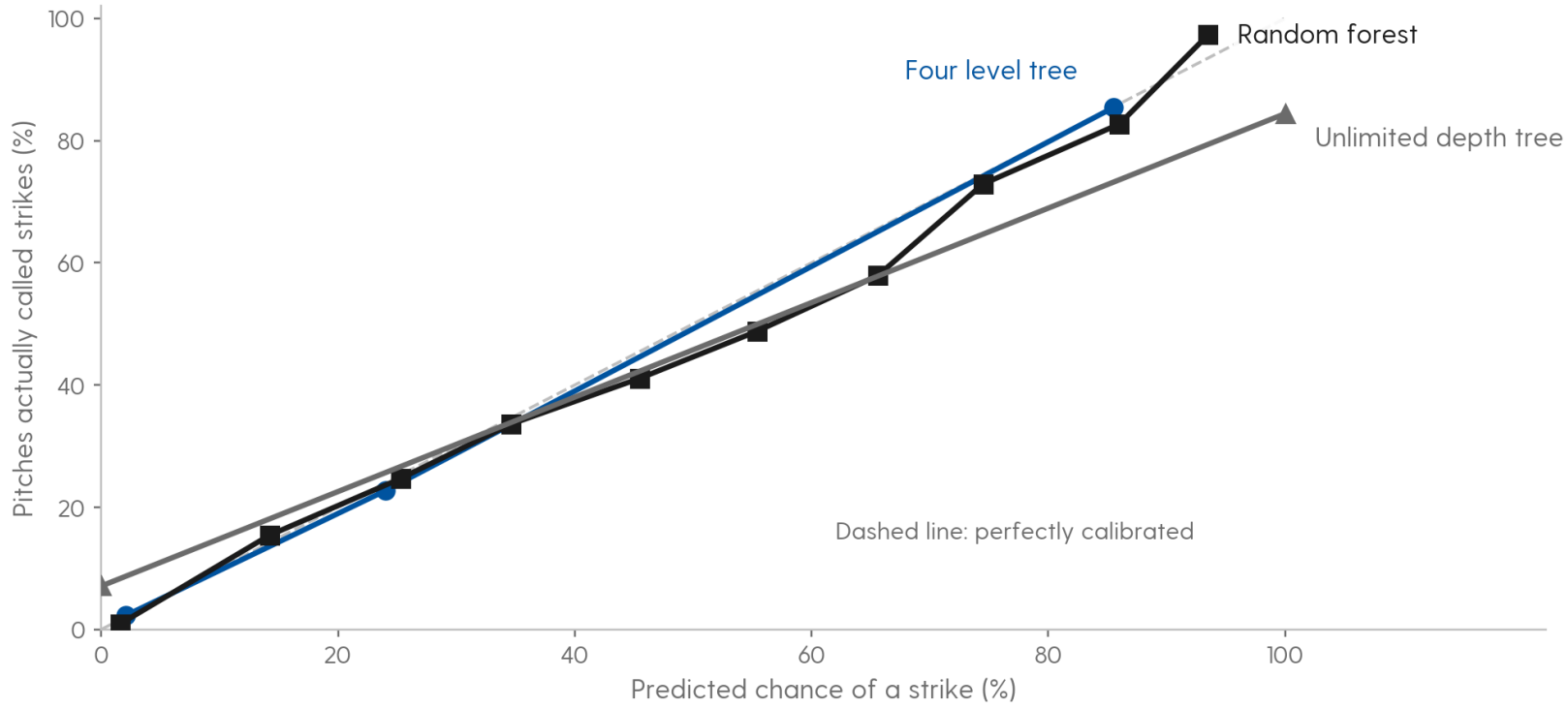
CALIBRATION

1. **Probabilities should mean what they say.**
   Of the pitches a model calls 70% strikes, about 70% should be strikes.

2. **Accuracy only checks which side of 50% a prediction lands on.**
   Calibration checks the number itself.

3. **Coaches act on the number.**
   Go for it at 60%? Kick at 55%?

CALIBRATION CHART



The four level tree and the forest sit near the line. The unlimited tree only says 0% or 100%, and its 100% pitches are strikes 84.4% of the time.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

CALIBRATION

BRIER SCORE

	ACCURACY	BRIER SCORE
League strike rate (31.7%) for every pitch	68.3%	0.217
Unlimited tree	90.2%	0.098
Four level tree	92.3%	0.066
Random forest, 8 levels	93.0%	0.050

Brier score: the average squared gap between the predicted probability and what happened. 0 is perfect; lower is better. The forest is best on both.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.



A MODEL SAYS THIS 4TH AND 1 CONVERTS 70% OF THE TIME. WHAT WOULD YOU NEED TO SEE BEFORE A COACH USES THAT NUMBER?

06

DATA LEAKAGE

DATA LEAKAGE

Using information that wouldn't be known when the prediction is made.

IN WEEK 5, WHY DID OPPONENTS' FULL-SEASON WIN PERCENTAGE PREDICT SECOND-HALF RECORDS SO WELL?



TOO GOOD TO BE TRUE?

4TH AND GOAL PROMPT

An NFL head coach asks what leads to success on 4th and goal from the 1. PFF data.

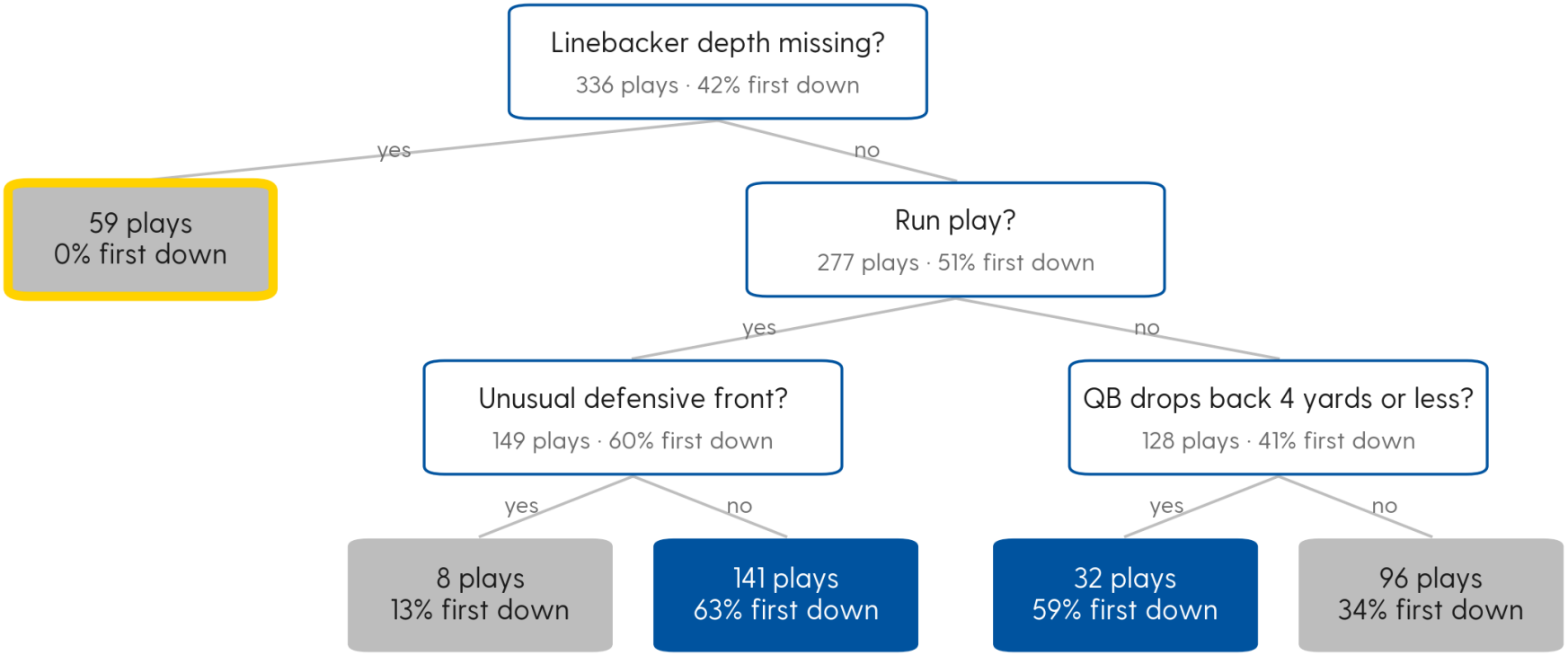
Target: pff_FIRST_DOWN_GAINED (1 = success, 0 = not).

Model: decision tree, max depth 3, 80/20 train/test split.

One-hot encode the categorical columns.

Optimize for recall; show the confusion matrix and the tree.

4TH AND GOAL DECISION TREE



The first split is whether linebacker depth is missing. All 59 plays where it was missing failed.

PFF play data, 2025 example.



WHY WOULD EVERY PLAY MISSING LINEBACKER DEPTH FAIL?

CHECKING FOR LEAKAGE

1. Make sure that every input is known at the decision point.
2. Be suspicious of a model that is nearly perfect.
3. Look at which variables have the most importance.

07

PROMPTING THE AGENT

PROMPT ENGINEERING

1. **Your question and audience.**

   A head coach wants to know what factors lead to success on 4th and 1 to help him decide what to do when he goes for it.

2. **How you define success. Ask an LLM to grill you on edge cases.**

   A first down. (What about a touchdown? What about a defensive penalty? Offensive penalty?)

3. **What data you have available and want to use.**

   I am interested in how defensive scheme, offensive formation, and run/pass decisions all play into this. Only include NFL regular season and playoff games. I do not want personnel ratings incorporated in the analysis.

4. **What you want to optimize for.**

   I want to predict if a 4th down is successful or not as accurately as possible. (Not precision or recall).

5. **What to watch out for.**

   Teams may go for it on 4th and short more often when they're losing. We need to control for offensive and defensive strength.

VOICE MEMO CONTEXT

Okay, 4th and 1. I keep going back and forth on it. I trust the sneak, our guy's good at it. But then you see a heavy front, they stack eight, nine guys in there, and our line just, we don't move anybody. That's what kills me. Going for it at our own 35 and getting stuffed, handing them a short field. I'd rather punt than do that. And does it even matter where we are on the field? Is the 40 different from the goal line? Oh, and the ones where there's a flag and the snap doesn't count, I don't care about those. Half the time the offense is doing a hard count just trying to draw the other team offsides. I just need something I can use on the sideline.

Fictional head coach transcription.

/GOAL

/goal <condition>

Sets a finish line, and Claude starts working toward it.

A second model checks after every turn.

A small, fast model reads the conversation and decides whether the condition is met.

It can’t run code or open files.

Write a condition Claude’s own output will show.

There is no built-in turn or time limit.

Add one: “or stop after 15 turns.”

/goal shows the status. /goal clear stops it.

VAGUE VS CONCRETE GOAL

VAGUE

/goal build a good 4th and 1 model

CONCRETE

/goal train a decision tree on 2016–2023 and test on 2024–2025; print accuracy, recall on failed attempts and the confusion matrix; list every column and confirm none is known only after the snap; or stop after 15 turns

08

THIS WEEK'S WORK

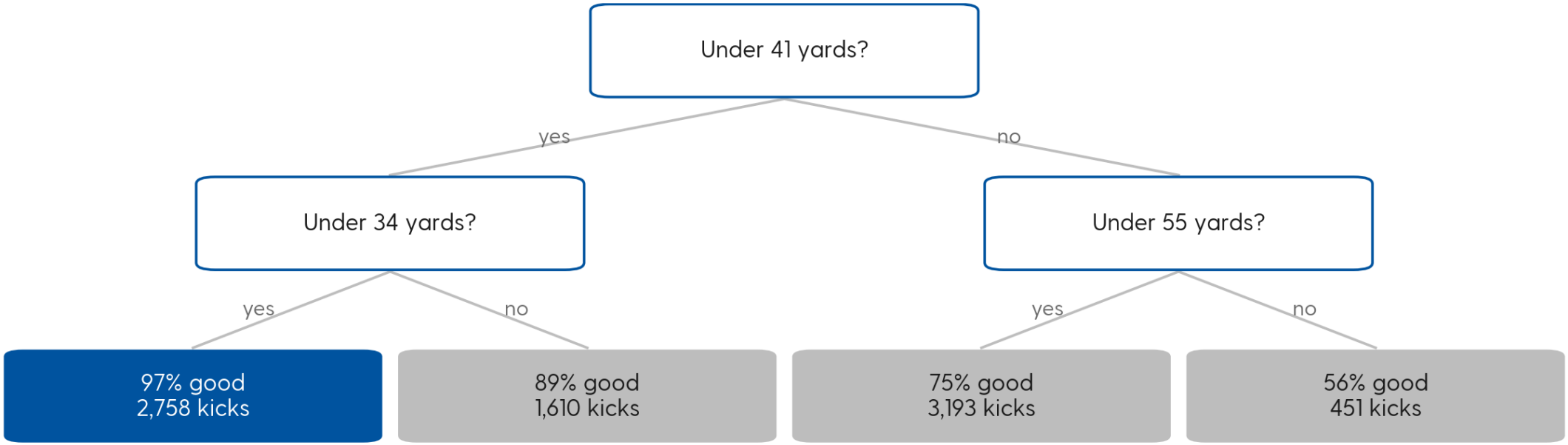
EVERY KICK IS GOOD

KICK DISTANCE	KICKS	MADE
Under 30 yards	2,311	97.7%
30 to 39 yards	2,865	92.3%
40 to 49 yards	2,966	78.4%
50 to 54 yards	1,291	70.3%
55 yards and up	582	56.0%

Saying every kick is good is right 84.8% of the time on the test kicks, and it never catches a miss.

Source: nflverse play-by-play, field goal attempts 2016 to 2025 (10,015 kicks).

TWO LEVEL TREE ON KICK DISTANCE



Every box is mostly makes, so it calls every kick good: 84.8% right, and it catches none of the misses.

Source: nflverse play-by-play, field goal attempts 2016 to 2025 (10,015 kicks).

A MISSED KICK COSTS MORE

	ACCURACY	MISSES CAUGHT	PREDICTED MISSES THAT WERE MISSES
Every mistake costs the same	84.8%	0.0%	None predicted
A missed kick costs more	67.9%	76.3%	28.9%

It predicts a miss from 38 yards out. It catches 76% of misses, and 29% of the kicks it calls misses really are.

Source: nflverse play-by-play, field goal attempts 2016 to 2025 (10,015 kicks).

THURSDAY

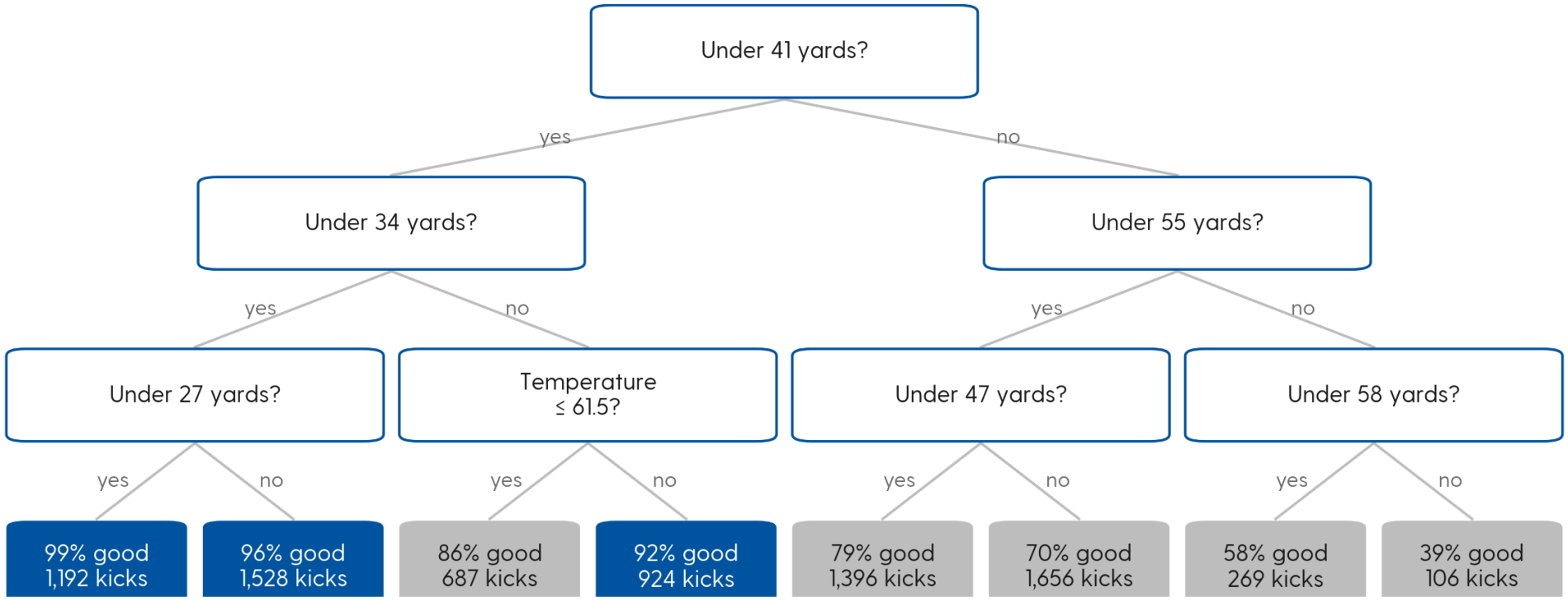
KICKER TREE WITH NO DEPTH LIMIT

	ACCURACY
Kicks it learned from	96.2%
New kicks	77.0%
Every kick is good, on new kicks	84.8%

96.2% on the kicks it learned from, 77.0% on new ones: worse than saying every kick is good.

Source: nflverse play-by-play, field goal attempts 2016 to 2025 (10,015 kicks).

THREE LEVEL FIELD GOAL TREE



Distance takes six of the seven questions. From 34 to 40 yards, outdoors at 61°F or colder, 86.0% are good; warmer or indoors, 92.0%.

Source: nflverse play-by-play, field goal attempts 2016 to 2025 (10,015 kicks).

FIELD GOAL TREE VS RANDOM FOREST

2024 AND 2025, 2,257 KICKS	ACCURACY	BRIER SCORE
Every kick good	84.9%	0.128
Three level tree	84.4%	0.118
Random forest	84.9%	0.116

Neither beats saying every kick is good on accuracy. On Brier score both do, and the forest edges the tree.

Source: nflverse play-by-play, field goal attempts 2016 to 2025 (10,015 kicks).



WHICH OF THESE WOULD YOU HAND A COACH, AND WHY?

YOUR TURN

Build the best field goal make-probability model you can.

Train on 2016 to 2023, test on 2024 and 2025.

Set a /goal first.

A concrete success condition and a turn cap. Save it as outputs/lab/goal.md and commit it before you run it.

Report on the test kicks.

Accuracy next to saying every kick is good, the Brier score, and a calibration chart.

Write two sentences for a kicking coach.

Data: weeks/week12/data/nfl-field-goals-2016-2025.csv after /update: 10,015 field goal attempts, 2016 to 2025.



WHICH RESULT SURPRISED YOU MOST, AND WHAT HAD YOU ASSUMED?



YOUR MODEL'S ACCURACY WENT DOWN AND YOU KEPT IT ANYWAY. WHY?

? WHAT DID CLAUDE GET WRONG TODAY, AND HOW DID YOU CATCH IT?

CASE STUDY

4TH & 1

STEPS

1. Commit your guess before you open the data: which play call converts most often, and about how often.
2. Write a prompt that covers the five parts on the Prompt engineering slide, and commit it before you run anything.
3. Build a decision tree, two or three levels, 0.8/0.2. Compare it with saying every play converts.
4. Check its probabilities: a calibration chart of the test plays, and the Brier score next to the league conversion rate for every play.
5. Read the tree and check each split against the plays.
6. Build a page where the coach picks a situation and sees the prediction.

BRIEF QUESTIONS

1. What would you tell your coach to call on 4th & 1, and what in your model supports it?
2. Was any column in the data something your coach couldn't know before the snap? How did you find out, and what did it do to your model?
3. Which mistake matters more to your coach here, a false positive or a false negative? How did that change the model you built?
4. What other data would you want before telling your coach to change his 4th & 1 calls?

Data: 2,689 plays, 2016 to 2025. Written to the head coach.

BEFORE YOU PUSH ON MONDAY, NOVEMBER 16

Commit goal.md, your lab model and calibration chart, prediction.md, prompt.md, your code, and your pages.

Answer every question in BRIEF.md, in your own words.

Run /audit, then /quiz-me.

Upload your repository's link to Canvas by Monday, November 16 at 11:59pm.